



Weekly report 2

Course

Master Thesis

Supervisor

Rasmus R. Paulsen

Paula Lopez Diez

Esther Creixell Mediante

Author

Aïda Jiménez Ordoñez

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1.1 Data available

The dataset used consists of anonymised head and neck CT scans acquired for radiotherapy purposes [1]. The scans were obtained from two available datasets hosted on The Cancer Imaging Archive [2]: the TCGA-HNSC dataset [3] and the Head-Neck Cetuximab dataset [4]. All selected cases correspond to CT planning studies with a uniform slice thickness of 2.5 mm, ensuring consistency across patients. In total, the dataset comprises 35 CT scans from 34 patients (9 of whom are from the TCGA-HNSC dataset and 25 from the Head-Neck Cetuximab dataset).

The patient cohort includes individuals diagnosed with head and neck cancer. Both the patient age and gender distribution of the 34 patients selected are shown in Figure 1 and Figure 2, respectively. Ethnicity is not reported.

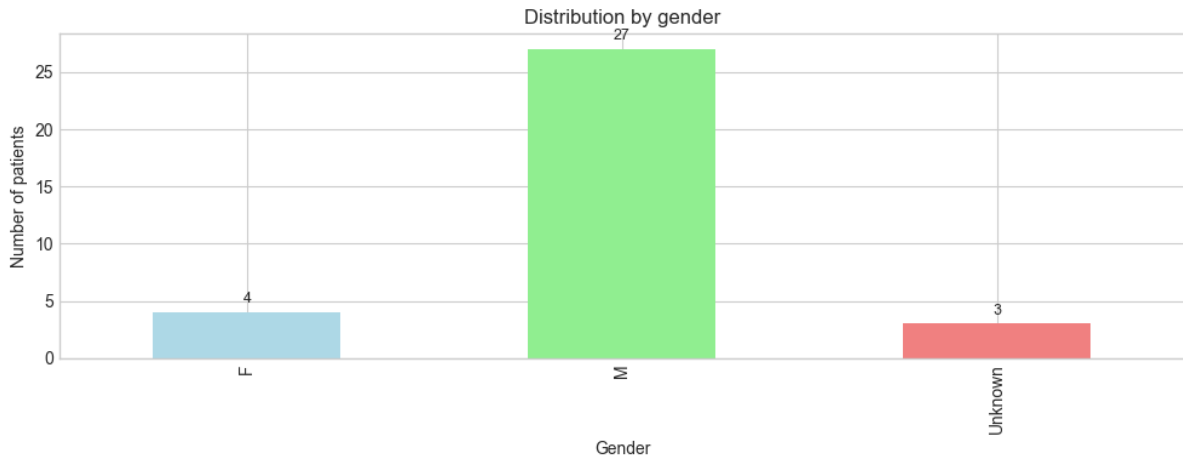


Figure 1: Gender distribution for the 34 patients selected.

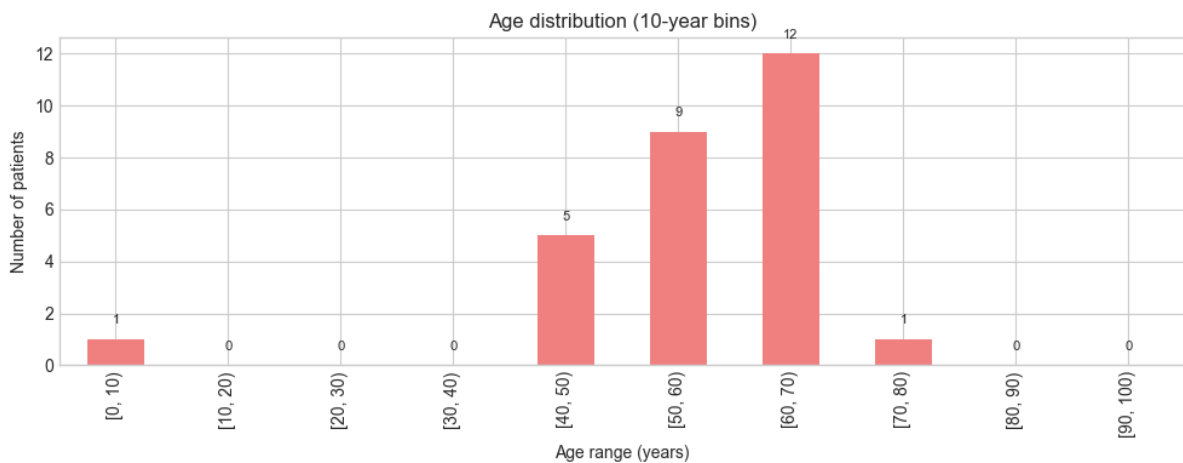


Figure 2: Age distribution for the 34 patients selected.

As observed, the data are unbalanced with respect to gender, as 79% of the patients are male. Moreover, three patients have an unknown gender, and only 12% are female. Regarding age, most patients are between 40 and 80 years old, with one exception between 0 and 10 years old, which may mean that the age is unknown, and another above 70.

A representative visualization of a patient's left ear, visualized in 3DSlicer using the axial view, is shown in Figure 3.

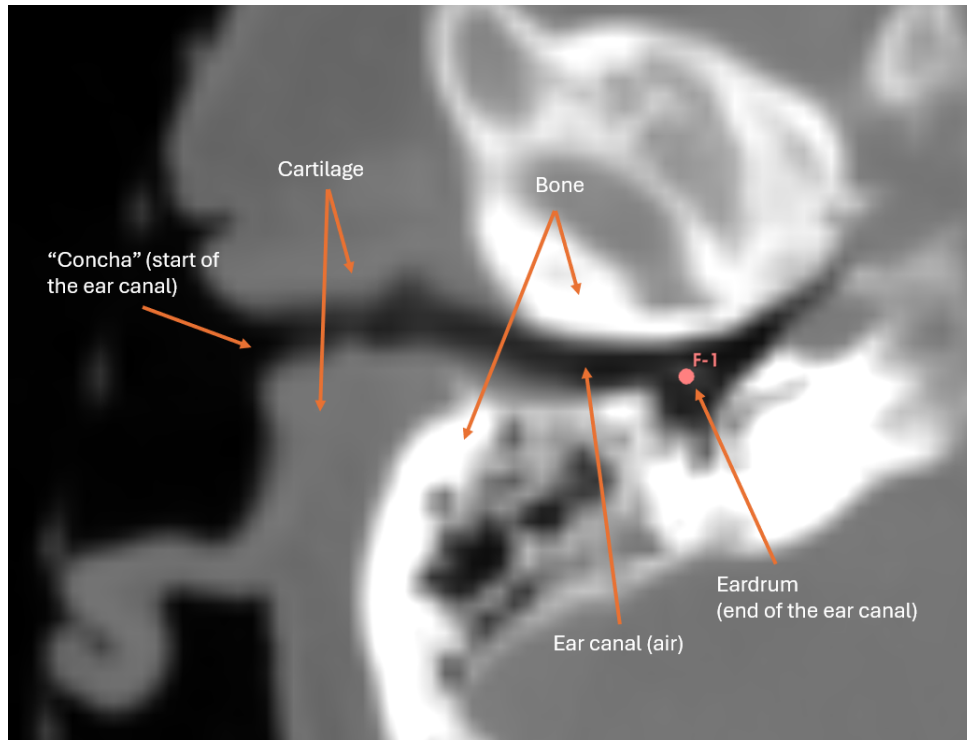


Figure 3: Example of one left ear visualized using 3DSlicer [5] axial view.

1.1.1 Data quality

All 35 CT scans were reviewed to verify data quality. While the majority focused on the head and neck, two scans consisted of full-body imaging (the two scans corresponding to the same patient) (see Figure 4a). Additionally, four scans exhibited superior truncation, where the top of the skull was cut off (see Figure 4b).

While the two CTs consisting of the entire body maintain a good quality and can be used, the four cases with the truncation should be omitted for the study as the entire head can be useful for further analysis.

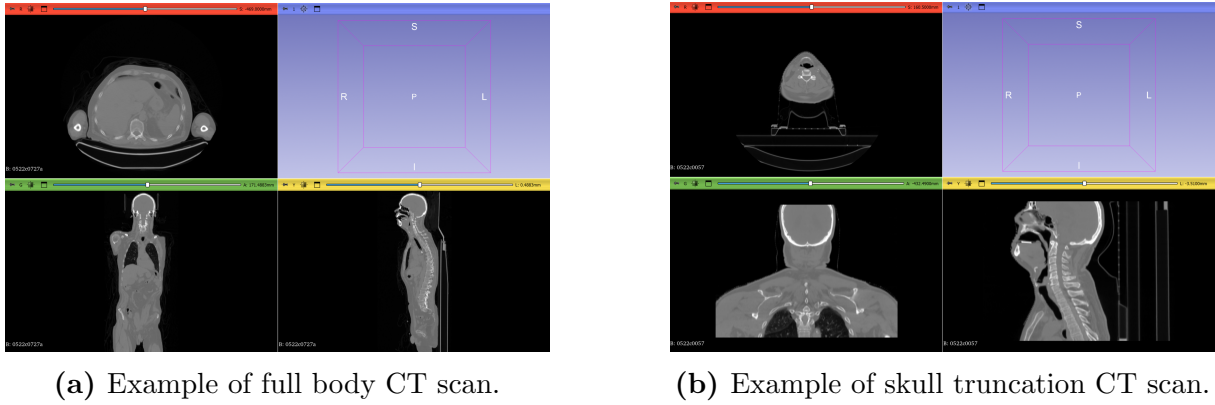


Figure 4: Data quality problem examples.

1.2 Segmentation

1.2.1 First approach: TotalSegmentator

As an initial approach, TotalSegmentator [6] was used to evaluate its performance in the automatic segmentation of ear and head-related anatomical structures.

TotalSegmentator is an open-source software framework designed for fully automated anatomical segmentation of medical imaging data, including both CT and MRI scans. It relies on deep learning techniques and provides pre-trained models capable of identifying and segmenting a large number of anatomical structures without requiring manual annotations. The framework is built on top of the nnU-Net architecture [7], which enables robust performance across heterogeneous datasets.

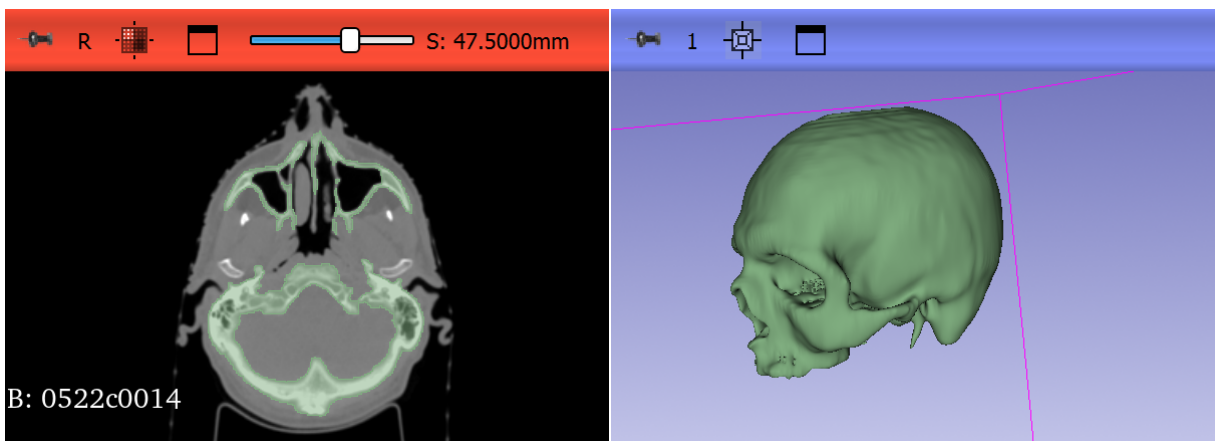


Figure 5: Example of skull segmentation obtained using the craniofacial bones model of TotalSegmentator.

Three different pre-trained models available within TotalSegmentator were explored to assess their suitability for segmenting anatomical regions relevant to the project. In particular, the *craniofacial bones* model was used to segment the skull and major bony structures of the head, as illustrated in Figure 5. Additionally, the same model was applied

to obtain a segmentation of the mandible (Figure 6) and the overall head anatomy, shown in Figure 7.

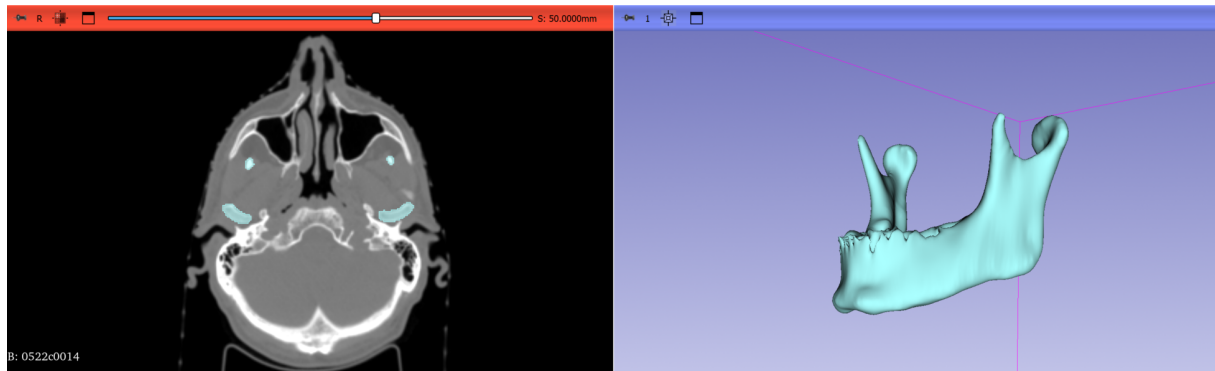


Figure 6: Example of mandible segmentation obtained using the craniofacial bones model of TotalSegmentator.

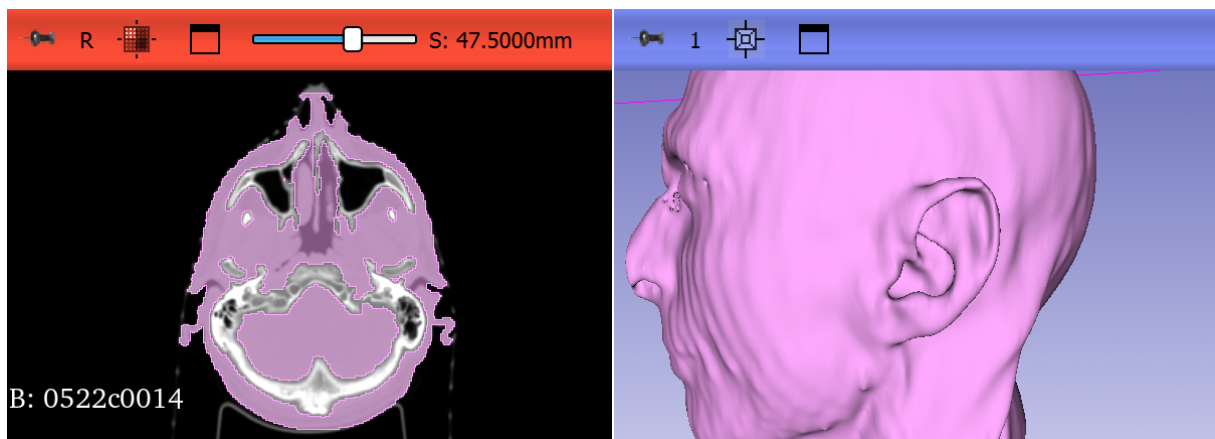


Figure 7: Example of head segmentation obtained using the craniofacial bones model of TotalSegmentator.

To specifically target ear-related structures, the *head gland cavities* model was applied. This model enabled the segmentation of the auditory canal for both left and right ears. An example of the resulting segmentation is shown in Figure 8.

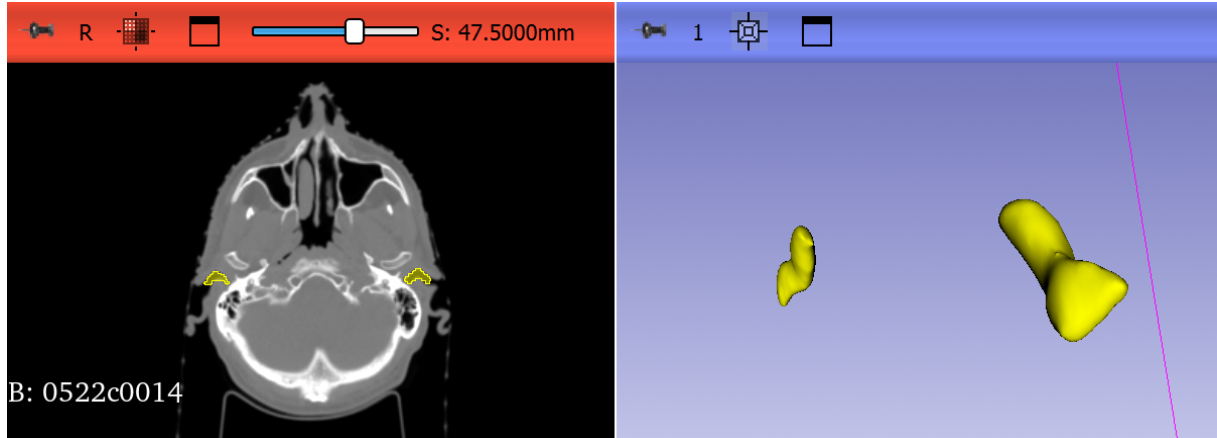


Figure 8: Example of auditory canal segmentation obtained using the head gland cavities model of TotalSegmentator.

Finally, the *head muscles* model was used to segment muscular structures of the head. In particular, the temporal muscle was successfully identified, as shown in Figure 9.

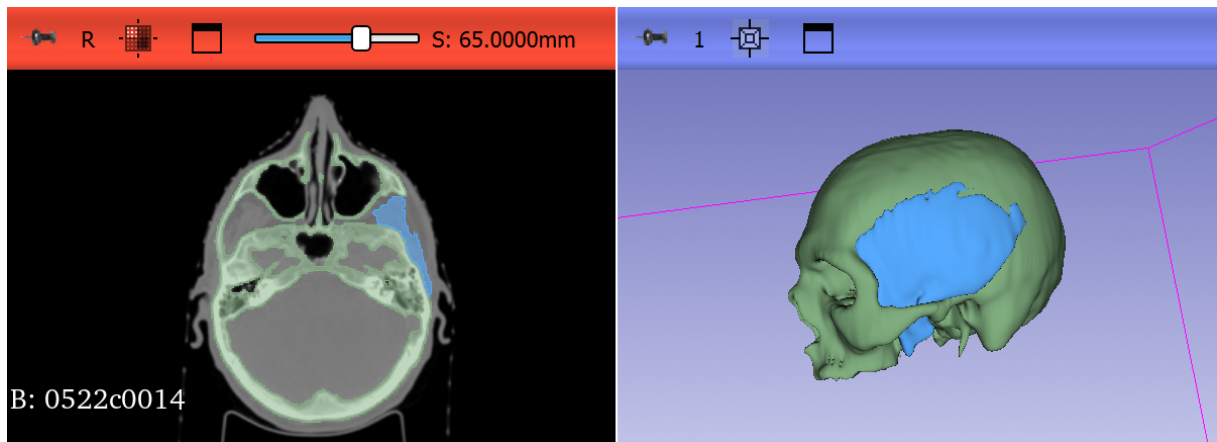


Figure 9: Example of temporal muscle segmentation obtained using the head muscles model of TotalSegmentator.

1.2.2 Thresholding-based segmentation

Thresholding-based segmentation is a classical image processing technique that separates objects from the background by classifying voxels according to their intensity values. In the context of CT imaging, this method relies on Hounsfield Unit (HU) values to distinguish between different tissue types. Voxels with intensities above or below a predefined threshold are assigned to a specific class, while the remaining voxels are excluded.

This approach is computationally efficient and easy to implement, making it suitable for rapid exploratory analysis. However, its effectiveness strongly depends on the presence of well-separated intensity distributions between the target structure and surrounding tissues. As a result, thresholding performs well for high-contrast structures such as bone, but is more challenging for soft tissues or anatomically complex regions where intensity overlap is significant.

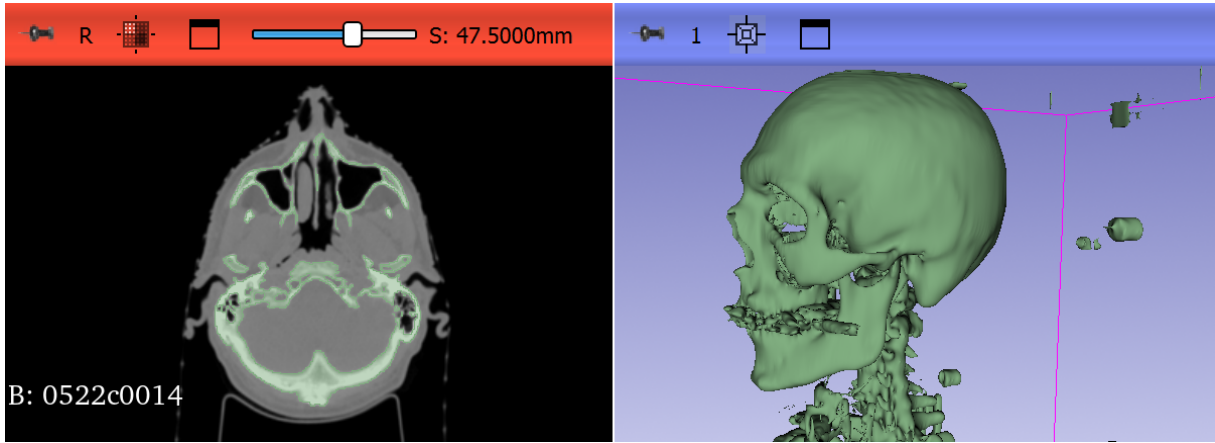


Figure 10: Skull segmentation obtained using thresholding with HU values above approximately 300 HU.

Figure 10 shows an example of bone segmentation obtained using a high-intensity threshold corresponding to cortical bone values (typically above 200-300 HU). In contrast, Figure 11 illustrates soft tissue segmentation using a lower HU range.

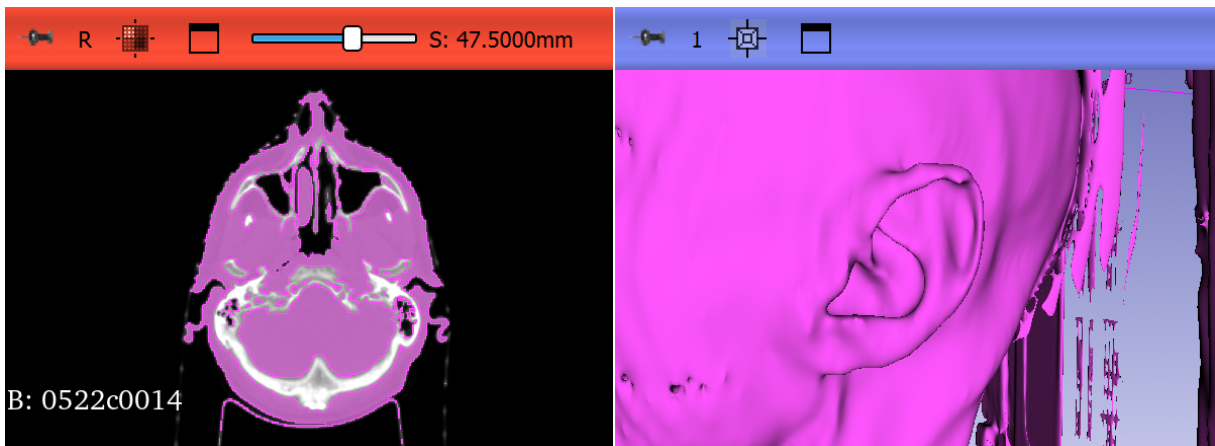


Figure 11: Soft tissue segmentation obtained using thresholding within an approximate HU range of -366 to 300 HU.

The same thresholding-based procedure was applied to segment the ear canal, using the low HU values corresponding to air (approximately -1000 HU). The resulting segmentation was subsequently refined through manual editing to improve anatomical accuracy (see Figure 12).

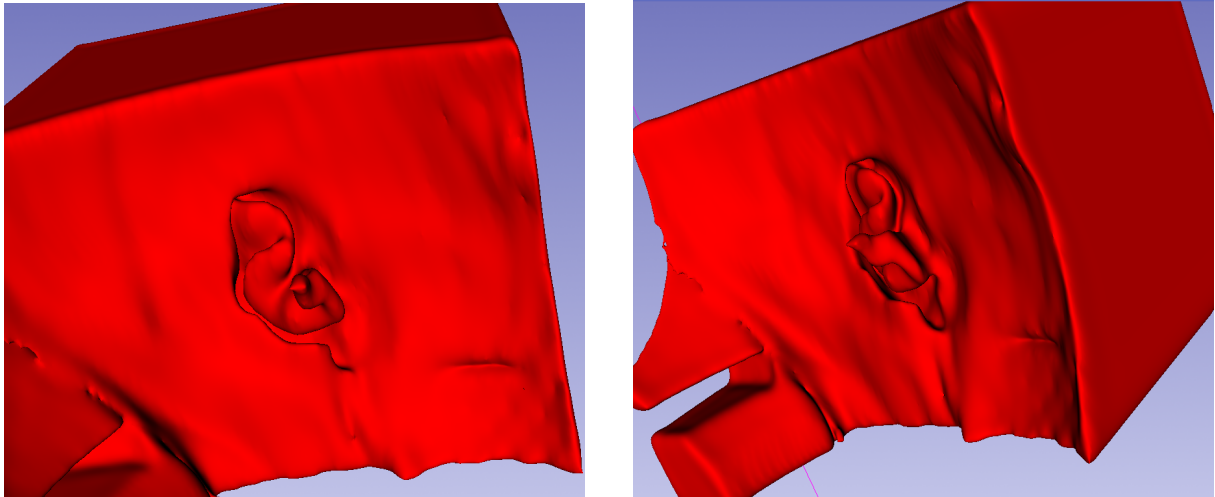


Figure 12: Example of ear canal segmentation obtained using the thresholding-based method.

1.2.3 Mix of both approaches + manual refinement

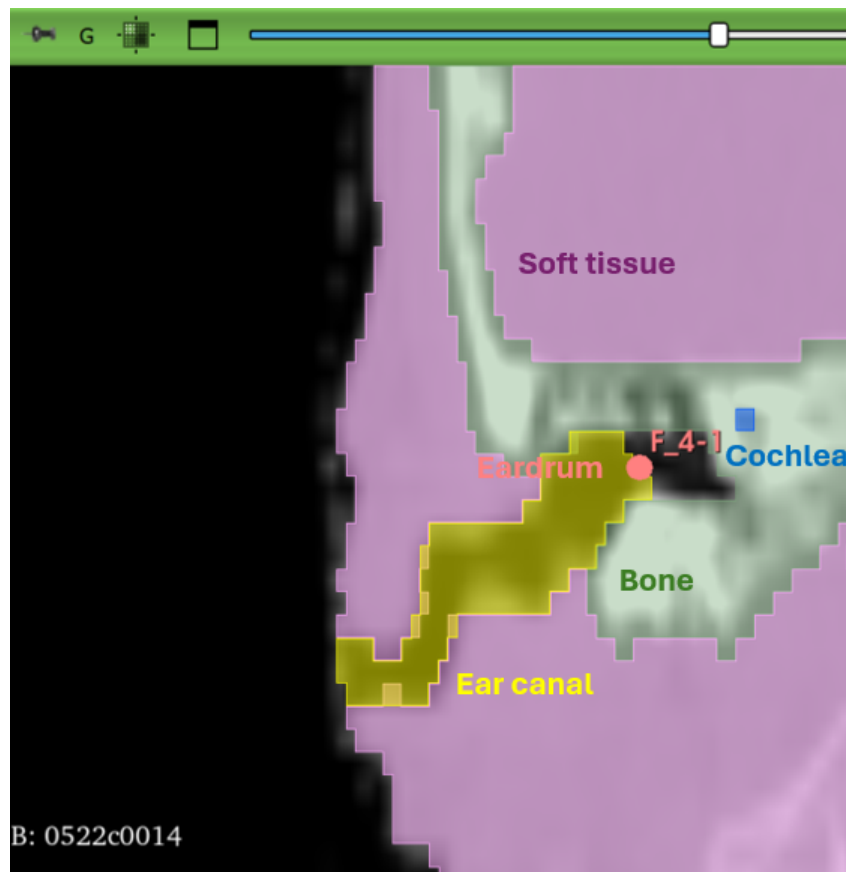


Figure 13: Combined preliminary segmentation showing soft tissue (pink), bone (green), air (yellow), cochlea (blue), and the approximate location of the tympanic membrane (red dot).

Figure 13 shows a preliminary combined segmentation obtained by overlaying multiple tissue classes derived from different segmentation approaches (TotalSegmentator and thresholding) and subsequently refined through manual editing. Soft tissue, bone, and air (ear canal) regions are simultaneously visualized (in the coronal plane), providing a comprehensive anatomical context of the ear region. Additionally, the approximate location of the tympanic membrane is indicated to highlight the end of the ear canal, as well as the segmentation of the cochlea.

1.2.4 Challenge

Differentiate between the various soft tissues: skin, cartilage, fat, and muscle (not visible in CT).

Possible idea: transfer knowledge from MRI images, where these soft tissues can be detected (to be defined).

1.2.5 General and initial approach

Step 1: thresholding/TS + manual correction to obtain the ground truth (GT) of the bone, soft tissue and air.

Step 2: create an AI-based model trained with these GTs.

(Step 3: use MRI to transfer soft tissue information)

Step 4: manually include the skin. Skin thickness covering the cartilage: 2 mm. Skin thickness covering bone: 1.5 mm [8].

1.3 Simulations Part: Theory

1.3.1 Finite Element Analysis

Finite Element Analysis (FEA) is a numerical method used to solve partial differential equations (PDEs) arising in mathematical physics and engineering. It represents a physical system by discretising a continuous domain Ω into a finite number of sub-regions, known as elements.

While analytical "hand calculations" are sufficient for simple geometries, real-world components exhibit complex behaviours described by the general equilibrium equation:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = \rho \ddot{\mathbf{u}} \quad (1)$$

where $\boldsymbol{\sigma}$ is the stress tensor, $\mathbf{f}$ is the body force, and $\mathbf{u}$ is the displacement field. FEA provides a framework to approximate these continuous fields.

1.3.2 Meshing

The core of FEA is the transformation of a continuous problem into a discrete one. This process, called meshing, replaces the irregular physical geometry with a collection of simple geometric shapes (e.g., hexahedrons or tetrahedrons) [9].

Each element is defined by a set of nodes. Within each element e , the unknown solution u (such as displacement or temperature) is approximated using shape functions N_i :

$$u^{(e)}(x, y, z) \approx \sum_{i=1}^n N_i(x, y, z) \hat{u}_i \quad (2)$$

where $\hat{u}_i$ represents the discrete values at the nodes.

The simulation process begins with the importation of a volumetric geometry. Because numerical solvers cannot directly compute irregular volumes, the mesh serves as the bridge between the design and the mathematical solution. High-quality meshing is essential to minimize discretization error and ensure the convergence of the solution [9].

1.3.3 The Mathematical Solver

The FEA solver assembles the individual element equations into a global system of linear algebraic equations. In structural mechanics, this typically takes the form:

$$\mathbf{K}\mathbf{U} = \mathbf{F} \quad (3)$$

where:

- $\mathbf{K}$ is the global stiffness matrix (representing material and geometry).
- $\mathbf{U}$ is the vector of unknown nodal displacements.
- $\mathbf{F}$ is the vector of external loads and boundary conditions.

1.3.4 Ansys Mechanical

Ansys Mechanical is a software application used for engineering simulation, particularly in structural analysis. It allows engineers to create detailed models of physical systems and test how they respond to forces, vibrations, heat, and other real-world conditions. By using FEA, Ansys Mechanical helps predict stress, deformation, and failure points in materials and components, enabling safer and more efficient designs [9].

1.4 TO DO

- Continue with the state-of-the-art.
- Try TS or thresholding with other CTs (just done with one case).
- Check how the soft tissue appears in MRI and if it is possible to use MRI for segmenting all the structures (including bone).
- Decide how to define the end of the ear canal: how to define or segment the eardrum.
- Develop an AI model to improve the segmentation outcomes. For this, I need to create ground-truth based on TS, thresholding and manual correction.

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